Md Al Amin Kazi

Postgraduate Diploma at IICT, BUET, Dhaka, Bangladesh
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RESEARCH INTERESTS

Optical Wireless Communication (LiFi, WiFi, 6G), Health AI, AI/ML, Computer Network, Cybersecurity

EDUCATION

WorldQuant University, Washington, D.C., USA (Online) — October 2025 — Expected June, 2027
Master of Science in Financial Engineering (MScFE) — Grade: 95%

Bangladesh University of Engineering and Technology, Dhaka, Bangladesh — July 2023 — Aug 2026
Postgraduate Diploma in ICT — CGPA: 3.40/4.00
Project Title: Machine Learning Techniques for Optimizing DCO-OFDM in LiFi Communication

Jashore University of Science and Technology, Jashore, Bangladesh — Jan 2017 — Dec 2020
Bachelor of Science in Physics — CGPA: 2.65/4.00
Project Title: An AB Initio Study of Structural, Electronic, and Optical Properties of Boron-Carbon-Nitride Bilayers

RESEARCH EXPERIENCE

Postgraduate Diploma Project — IICT, BUET
Machine Learning Techniques for Optimizing DCO-OFDM in LiFi Communication — 2026
Supervisor: Prof. Dr. Md. Rubaiyat Hossain Mondal — IICT, BUET, Dhaka, Bangladesh

- Simulated a 27,720-sample DCO-OFDM LiFi dataset across $M \in [4, 4096]$ and $N \in [64, 4096]$ in MATLAB.
- Benchmarked 42 regression models using LazyPredict; engineered HistGradientBoosting, MLP, and Polynomial Regression to find the optimal predictor.
- Established HGBRegressor as the top predictor using only two statistical features, reducing feature overhead by 60% and obtaining an R-squared score of 0.9958 and an RMSE of 0.0575.

ELITE Research Lab LLC — Dhaka, Bangladesh
Research Assistant — July 2026 — Present
Project: A cross-layer, multi-cell-aware handover and resource-allocation framework for hybrid LiFi-WiFi networks that jointly optimizes communication throughput and sensing accuracy via deep reinforcement learning.
Supervisor: Md Kishor Moral (PhD, Cornell University) — Founder & CEO, ELITE Research Lab LLC; former Adjunct Faculty, St. Thomas University

- Designing an RF/LiFi handover model to preserve seamless connectivity and continuous sensing.
- Creating a multi-cell architecture that suppresses communication interference while leveraging inter-cell signals for sensing, leading to seamless communication and sensing.
- Implementing a Deep Reinforcement Learning system for resource allocation to jointly optimize network throughput and sensing accuracy.

Medora Research Institute - MRI — Dhaka, Bangladesh
Research Assistant — Oct 2025 — Present
Project 1: Explainable AI-Based Cancer Prediction Using Advanced Visualization and Machine Learning Models.
Supervisor: Dr. Md. Murad Hossain — Lecturer, Dept. of Statistics, Gopalganj Science and Technology University

- Built a leak-free cancer prediction pipeline using XGBoost, Random Forest, MLP, and Logistic Regression with grid search and 5-fold CV.
- Achieved top performance with XGBoost: 93.67% accuracy, 0.9663 ROC-AUC, and 0.9564 mean CV AUC.
- Applied SHAP, LIME, and statistical effect-size analysis (Cohen's d, odds ratios) to extract key risk and protective clinical drivers.

Project 2: A Machine Learning Framework for Heart Attack Risk Prediction Using Multidimensional Health Data with Explainable AI
Supervisor: Dr. Md Murad Hossain — Lecturer, Dept. of Statistics, Gopalganj Science and Technology University

- Built an end-to-end heart attack risk prediction pipeline on 8,763 instances with 26 clinical, demographic, behavioral, and geographical features.
- Evaluated Logistic Regression, Random Forest, and Histogram Gradient Boosting (HistGB accuracy = 0.633, ROC-AUC = 0.530, leading to relatively poor metrics, but the subgroup analysis and XAI backed it for holding the standard.
- Applied SHAP, LIME, permutation importance, calibration analysis, threshold tuning, and demographic subgroup evaluations.

PUBLICATIONS

Journal paper
Submitted:

- Kazi, M. A. A., and Mondal M. R. H.*. Machine Learning Techniques for Optimizing DCO-OFDM in LiFi Communication.
- Arman, M., Shuba, S. J., Ahamed, M. S., Hossain, M. A., Kazi, M. A. A., Abdullah, S., Dohan, D. M., Zuberi, N.*. Automated Fracture Image Captioning Using Multimodal Vision-Language Models: A Comprehensive Comparative Study on a Clinically Curated Dataset.
- Fariba, M. H., Brinto, M. F., Kazi, M. A. A., Ahmed, M. T., Khan, M. R., Islam, M. M., Sorna, S. S., Siam, N. F., Hossain, M. M.*. Explainable AI-Based Cancer Prediction Using Advanced Visualization and Machine Learning Models.
- Paul, S., Kazi, M. A. A., Saidunnessa, Roy, D., Deb, S., Refat, M. K., Islam, M. N., Faiyaz, Hossain, M. M.*. A Machine Learning Framework for Heart Attack Risk Prediction Using Multidimensional Health Data with Explainable AI.

SELECTED COURSES

Postgraduate Diploma Courses

- Data Communications
- Introduction to Telecommunications
- Embedded System Design
- Computer Networks
- Data Structure and Algorithm
- Software Engineering and Application Development
- Visual Programming
- Programming Concept
- Database Design and Management
- Operating System Concepts

Additional Courses

- Supervised Machine Learning: Regression and Classification
- Advanced Learning Algorithms
- Data Analysis with Python
- How Google does Machine Learning
- Introduction to Machine Learning
- Machine Learning with Python (V2)
- Python Essentials for MLOps
- Introduction to Programming
- Power BI for Business Professionals
- UNDERSTANDING THE INTERNET (BLENDED LEARNING)

Bachelor's Courses

- Optics
- Lasers and Fiber Optic Communications
- Vibrations and Waves
- Electricity and Magnetism
- Electronics I
- Electronics II
- Classical Electrodynamics I
- Classical Electrodynamics II
- Medical and Radiation Physics
- Principles of Statistics
- Computational Physics
- Mathematical Physics
- Numerical Methods
- Differential and Integral Calculus

Workshop and Conference

Inclusive Education, DiversAsia Toolkit and Accessible Reading Materials — Dhaka, Bangladesh
Conducted by BUET — 18 July, 2023

DiversAsia International Conference on Inclusive Higher Education: Bangladesh Context — Dhaka, Bangladesh
Held at BUET — 20-21 Sep, 2023

ENGLISH TEST

IELTS (Academic): 7.5 (overall score)

Listening: XX — Reading: XX
Speaking: XX — Writing: XX
Test date: August 2026

SKILLS

- **Programming:** Python, Java, C, SQL, MATLAB
- **AI/ML (Frameworks):** PyTorch, TensorFlow, Keras
- **AI/ML (Libraries):** Scikit-learn, NumPy, Pandas, SciPy, Matplotlib, Seaborn
- **AI/ML (Tools) :** Google Colab, NotebookLM, Git, LaTeX, Google Cloud Platform

REFERENCES

Prof. Dr. Md. Rubaiyat Hossain Mondal
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Scholar Profiles: BUET Profile — Google Scholar — ResearchGate

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